

10

Experimental tests of general relativity

Most of the experimental tests of general relativity are based on the Schwarzschild geometry in the region $r > 2GM/c^2$. Some are based on the trajectories of massive particles and others on photon trajectories. Most of the ‘classic’ tests are in the weak-field limit, but more recent observations have begun to probe the more discriminative strong-field regime. We will now discuss both these ‘classic’ experimental tests and some of the more recent findings and proposals. Some later tests are in fact more closely linked to the Kerr geometry (see Chapter 13), which describes spacetime outside a rotating massive body, but the basic principles can still be understood in terms of the simpler Schwarzschild geometry.

10.1 Precession of planetary orbits

For a general non-circular orbit in Newtonian theory the equation of motion is

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{h^2},$$

where $u \equiv 1/r$ and h is the angular momentum per unit mass of the orbiting particle. For a bound orbit, the equation has the solution

$$u = \frac{GM}{h^2}(1 + e \cos \phi), \quad (10.1)$$

which describes an ellipse; the parameter e measures the *ellipticity* of the orbit. Thus, for example, we can draw the orbit of a planet around the Sun as in Figure 10.1. We can write the distance of closest approach (*perihelion*) as $r_1 = a(1 - e)$ and the distance of furthest approach (*aphelion*) as $r_2 = a(1 + e)$. The equation of motion then requires that the semi-major axis is given by

$$a = \frac{h^2}{GM(1 - e^2)}. \quad (10.2)$$

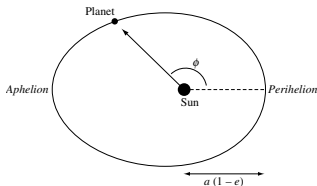


Figure 10.1 The elliptical orbit of a planet around the Sun; e is the ellipticity of the orbit.

The general-relativistic equation of motion is

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{h^2} + \frac{3GM}{c^2} u^2. \quad (10.3)$$

If the gravitational field is weak, as it is for planetary orbits around the Sun, then we expect Newtonian gravity to provide an excellent approximation to the motion in general relativity. We can therefore treat the Newtonian solution (10.1) as the zeroth-order solution to the general-relativistic equation of motion. Thus let us write the general-relativistic solution as

$$u = \frac{GM}{h^2} (1 + e \cos \phi) + \Delta u,$$

where Δu is a perturbation. Substituting this expression into the general-relativistic equation (10.3) we find that, to first-order in Δu ,

$$\frac{d^2 \Delta u}{d\phi^2} + \Delta u = A (1 + e^2 \cos^2 \phi + 2e \cos \phi),$$

where the constant $A = 3(GM)^3/(c^2 h^4)$ is very small. A particular integral of this equation is easily found to be

$$\Delta u = A \left[1 + e^2 \left(\frac{1}{2} - \frac{1}{6} \cos 2\phi \right) + e\phi \sin \phi \right], \quad (10.4)$$

which can be checked by direct differentiation.

Since the constant A is very small, the first two terms on the right-hand side of (10.4) are tiny, and of no use in testing the theory. However, the last term, $Ae\phi \sin \phi$, might be tiny at first but will gradually grow with time, since the factor

ϕ means that it is *cumulative*. We must therefore retain it, and so our approximate solution reads

$$u = \frac{GM}{h^2} [1 + e(\cos \phi + \alpha \phi \sin \phi)], \quad (10.5)$$

where $\alpha = 3(GM)^2/(h^2 c^2) \ll 1$. Using the relation

$$\begin{aligned} \cos[\phi(1-\alpha)] &= \cos \phi \cos \alpha \phi + \sin \phi \sin \alpha \phi \\ &\approx \cos \phi + \alpha \phi \sin \phi \quad \text{for } \alpha \ll 1, \end{aligned} \quad (10.6)$$

we can therefore write

$$u \approx \frac{GM}{h^2} \{1 + e \cos[\phi(1-\alpha)]\}. \quad (10.7)$$

From this expression, we see that the orbit is periodic, but with a period $2\pi/(1-\alpha)$, i.e. the r -values repeat on a cycle that is larger than 2π . The result is that the orbit cannot 'close', and so the ellipse *precesses* (see Figure 10.2). In one revolution, the ellipse will rotate about the focus by an amount

$$\Delta\phi = \frac{2\pi}{1-\alpha} - 2\pi \approx 2\pi\alpha = \frac{6\pi(GM)^2}{h^2 c^2}.$$

Substituting for h from (10.2), we finally obtain

$$\Delta\phi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

(10.8)

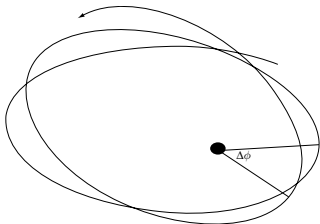


Figure 10.2 Precession of an elliptical orbit (greatly exaggerated).

Let us apply equation (10.8) to the orbit of Mercury, which has the following parameters: period = 88 days, $a = 5.8 \times 10^{10}$ m, $e = 0.2$. Using $M_{\odot} = 2 \times 10^{30}$ kg, we find

$$\Delta\phi = 43'' \text{ per century.}$$

In fact, the measured precession is

$$5599''.7 \pm 0''.4 \text{ per century,}$$

but almost all of this is caused by perturbations from other planets. The residual, after taking perturbations into account, is in remarkable agreement with general relativity. The residuals for a number of planets (and Icarus, which is a large asteroid with a perihelion that lies within the orbit of Mercury) may also be calculated (in arcseconds per century):

	Observed residual	Predicted residual
Mercury	43.1 ± 0.5	43.03
Venus	8 ± 5	8.6
Earth	5 ± 1	3.8
Icarus	10 ± 1	10.3

In each case, the results are in excellent agreement with the predictions of general relativity. Einstein included this calculation regarding Mercury in his 1915 paper on general relativity. He had solved one of the major problems of celestial mechanics in the very first application of his complicated theory to an empirically testable problem. As you can imagine, this gave him tremendous confidence in his new theory.

10.2 The bending of light

We have already noted that a massive object can have a significant effect on the propagation of photons. For example, photons can travel in a circular orbit at $r = 3GM/c^2$. We do not, however, expect to observe this effect directly, but a more modest bending of light can be observed. For investigating the slight deflection of light by, for example, the Sun, it is easiest to follow an approximation technique analogous to that used in predicting the perihelion shift of Mercury.

As we showed in Chapter 9, the 'shape' equation for a photon trajectory in the equatorial plane of the Schwarzschild geometry is

$$\frac{d^2u}{d\phi^2} + u = \frac{3GM}{c^2}u^2, \quad (10.9)$$

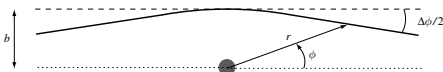


Figure 10.3 Angles and coordinates in the deflection of light by a spherical mass.

where $u \equiv 1/r$. In the absence of matter, the right-hand side vanishes and we may write the solution as

$$u = \frac{\sin \phi}{b}, \quad (10.10)$$

which represents a straight-line path with impact parameter b (see Figure 10.3). We again treat (10.10) as the zeroth-order solution to the equation of motion. Thus we write the general-relativistic solution as

$$u = \frac{\sin \phi}{b} + \Delta u,$$

where Δu is a perturbation. Substituting this expression into (10.9), we find that, to first order in Δu ,

$$\frac{d^2 \Delta u}{d\phi^2} + \Delta u = \frac{3GM}{c^2 b^2} \sin^2 \phi.$$

This is satisfied by the particular integral

$$\Delta u = \frac{3GM}{2c^2 b^2} \left(1 + \frac{1}{3} \cos 2\phi\right), \quad (10.11)$$

and adding (10.11) into the original solution yields

$$u = \frac{\sin \phi}{b} + \frac{3GM}{2c^2 b^2} \left(1 + \frac{1}{3} \cos 2\phi\right). \quad (10.12)$$

Now consider the limit $r \rightarrow \infty$, i.e. $u \rightarrow 0$. Clearly, for a slight deflection we can take $\sin \phi \approx \phi$ and $\cos 2\phi \approx 1$ at infinity, to obtain $\phi = -2GM/(c^2 b)$. Thus the total deflection (see Figure 10.3) is

$$\Delta \phi = \frac{4GM}{c^2 b}.$$

(10.13)

This is the famous gravitational deflection formula (which incidentally is twice what had previously been worked out using a Newtonian approach). For light

grazing the Sun it yields $\Delta\phi = 1''.75$. The 1919 eclipse expedition led by Eddington gave two sets of results:

$$\Delta\phi = 1''.98 \pm 0''.16,$$

$$\Delta\phi = 1''.61 \pm 0''.4,$$

both consistent with the theory. This provided the first experimental verification of a *prediction* of Einstein's theory (the 'anomalous' perihelion shift of Mercury had been known for many years) and turned Einstein into a scientific superstar.¹ Some historians have argued that Eddington 'fiddled' the results to agree with the theory. If Eddington did indeed massage the results, then he gambled correctly. Later high-precision tests using *radio sources*, which can be observed near the Sun even when there is no lunar eclipse, show there is now no doubt that the general-relativistic prediction is accurate to a fraction of a percent. Modern radio experiments using very long baseline interferometry (VLBI) have been performed to measure the gravitational deflection of the positions of radio quasars as they are eclipsed by the Sun. Such experiments can be performed to an accuracy of better than $\sim 10^{-4}$ arcseconds. Figure 10.4 summarizes the results of measurements of the deflection angle $\Delta\phi$ from experiments conducted over the period 1969–75. The results are in excellent agreement with the predictions of general relativity. Moreover, as one can see from the figure, the results constrain the parameter ω in the Brans–Dicke theory of gravity (see Appendix 8A): we have $\omega \geq 40$.

For more dramatic light deflection, our adopted approach of successive approximations is unsuitable. In this case, it is more appropriate to use the exact equation for $d\phi/dr$ derived in Chapter 9, which reads

$$\frac{d\phi}{dr} = \frac{1}{r^2} \left[\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2\mu}{r} \right) \right]^{-1/2},$$

where b is the impact parameter at infinity. We also showed in Chapter 9 that if $b > 3\sqrt{3}\mu$ then the photon is not captured by the mass; the resulting general orbit shape is illustrated in Figure 10.5. From the figure, we see that the deflection angle is given by

$$\Delta\phi = 2 \int_{r_0}^{\infty} \frac{1}{r^2} \left[\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2\mu}{r} \right) \right]^{-1/2} dr \quad (10.14)$$

where r_0 is the point of closest approach, at which the expression in the square brackets in (10.14) vanishes.

¹ The media had a great story. Remember that this was just after the end of the First World War, and so the headlines read something like 'Newton's theory of gravity overthrown by German physicist, verified by British scientists.'

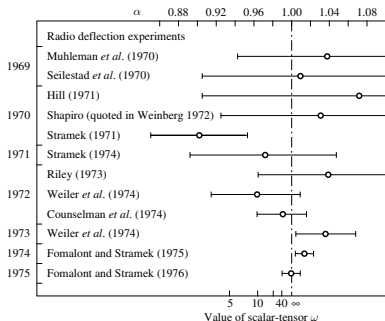


Figure 10.4 Results of radio-wave deflection measurements of the positions of quasars in the period 1969–75 (from C. Will, *Theory and Experiment in Gravitational Physics*, Cambridge University Press, 1981). The deflection angle is $\Delta\phi = \alpha 4GM/(R_{\odot}c^2)$ and the error bars are plotted on the parameter α . If general relativity is correct, we expect $\alpha = 1$. The abscissa scale gives the measured values of the parameter ω in the Brans–Dicke scalar-tensor theory of gravity, discussed in Appendix 8A.

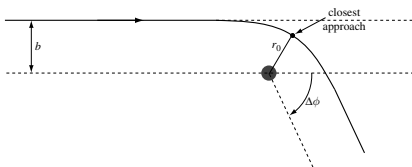


Figure 10.5 Angles and coordinates in the deflection of light by a spherical mass.

10.3 Radar echoes

In Chapter 9, we showed that the ‘energy’ equation for a photon orbit in the Schwarzschild geometry is

$$\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2\mu}{r} \right) = c^2 k^2.$$

Using the result

$$\left(\frac{dr}{d\tau}\right)^2 = \left(\frac{dr}{dt} \frac{dt}{d\tau}\right)^2 = \frac{k^2}{(1-2\mu/r)^2} \left(\frac{dr}{dt}\right)^2,$$

we can rewrite the energy equation as

$$\frac{1}{(1-2\mu/r)^3} \left(\frac{dr}{dt}\right)^2 + \frac{h^2}{k^2 r^2} - \frac{c^2}{1-2\mu/r} = 0. \quad (10.15)$$

Now consider a photon path from Earth to another planet (say Venus), as shown in Figure 10.6. Evidently the photon path will be deflected by the gravitational field of the Sun (assuming that the planets are in a configuration like that shown in the figure, where the photon has to pass close to the Sun in order to reach Venus). Let r_0 be the coordinate distance of closest approach of the photon to the Sun; then

$$\left(\frac{dr}{dt}\right)_{r_0} = 0,$$

and so from (10.15) we have

$$\frac{h^2}{k^2 r_0^2} = \frac{c^2}{1-2\mu/r_0}.$$

Thus, after rearrangement, we can write (10.15) as

$$\frac{dr}{dt} = c(1-2\mu/r) \left[1 - \frac{r_0^2(1-2\mu/r)}{r^2(1-2\mu/r_0)} \right]^{1/2},$$

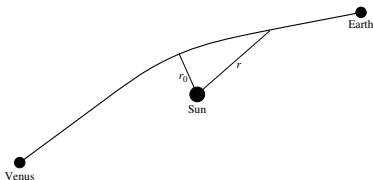


Figure 10.6 Photon path from Earth to Venus deflected by the Sun.

which can be integrated to give for the time taken to travel between points r_0 and r

$$t(r, r_0) = \int_{r_0}^r \frac{1}{c(1-2\mu/r)} \left[1 - \frac{r_0^2(1-2\mu/r)}{r^2(1-2\mu/r_0)} \right]^{-1/2} dr.$$

The integrand can be expanded to first order in μ/r to obtain

$$t(r, r_0) = \int_{r_0}^r \frac{r}{c(r^2 - r_0^2)} \left[1 + \frac{2\mu}{r} + \frac{\mu r_0}{r(r+r_0)} \right] dr,$$

which can be evaluated to give

$$t(r, r_0) = \frac{(r^2 - r_0^2)^{1/2}}{c} + \frac{2\mu}{c} \ln \left[\frac{r + (r^2 - r_0^2)^{1/2}}{r_0} \right] + \frac{\mu}{c} \left(\frac{r - r_0}{r + r_0} \right)^{1/2}. \quad (10.16)$$

The first term on the right-hand side is just what we would have expected if the light had been travelling in a straight line. The second and third terms give us the extra coordinate time taken for the photon to travel along the *curved* path to the point r . So, you can see from Figure 10.6 that if we bounce a radar beam to Venus and back then the excess coordinate-time delay over a straight-line path is

$$\Delta t = 2 \left[t(r_E, r_0) + t(r_V, r_0) - \frac{(r_E^2 - r_0^2)^{1/2}}{c} - \frac{(r_V^2 - r_0^2)^{1/2}}{c} \right],$$

where the factor 2 is included because the photon has to go to Venus and back. Since $r_E \gg r_0$ and $r_V \gg r_0$ we have

$$t(r_E, r_0) - \frac{(r_E^2 - r_0^2)^{1/2}}{c} \approx \frac{2\mu}{c} \ln \left(\frac{r_E}{r_0} \right) + \frac{\mu}{c},$$

and likewise for t_V and r_V . Thus, the excess coordinate-time delay is

$$\Delta t \approx \frac{4GM}{c^3} \left[\ln \left(\frac{r_E r_V}{r_0^2} \right) + 1 \right].$$

Of course, clocks on earth do not measure coordinate time but the corresponding proper time. Assuming the Earth to be at fixed coordinates (r, θ, ϕ) during the travel time of the signal, this is given by

$$\Delta \tau = \left(1 - \frac{2GM}{c^2 r_E} \right)^{1/2} \Delta t.$$

However, since $r_E \gg GM/c^2$, we can ignore this effect to the accuracy of our calculation. For Venus, when it is opposite to the Earth on the far side of the Sun,

$$\Delta \tau \approx 220 \mu s.$$

The idea of the experiment is as follows. Fire an intense radar beam towards Venus when it is almost opposite to the Earth on the far side of the Sun and measure the time delay of the radar echo with a sensitive radio telescope. The excess time delay gives us a test of the principle of equivalence. This sounds straightforward, but the time delay is very small and depends on the values of r_E , r_V , r_0 . How can one determine these parameters to the required precision? The answer is to fit the measured delays over a long period of time to a curve chosen by varying r_E , r_V , μ etc. as free parameters (see Figure 10.7). There are a number of technical problems that limit the accuracy of this method. Firstly, we must correct for the motion of Venus and the Earth in their orbits and for their individual gravitational fields. Also, in practice, the radar beam is reflected from different points on the surface of Venus (mountain peaks, valleys, etc.) and this introduces a dispersion in the time delay of several hundred μ s. This problem can be solved by bouncing the radar beams from a mirror – as has since been done using the Viking landers on Mars. Another, more complicated, problem is correcting for refraction by the Solar corona – this can be important for photon paths that graze the surface of the Sun. Nevertheless, Figure 10.7 confirms that the corrected measurements are in excellent agreement with the general-relativistic prediction.

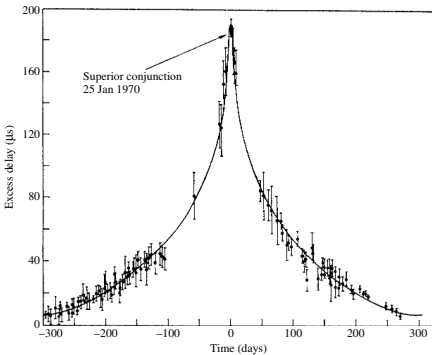


Figure 10.7 The Earth–Venus time-delay measurement compared with the general-relativistic prediction.

10.4 Accretion discs around compact objects

As we have seen, the orbits of particles and photons are probes of the geometry of spacetime. Information about the geometry produced by compact massive objects or black holes can be obtained from observations of the orbits of particles in the accretion disc that often surrounds them. As we saw in Chapter 9, the radiation efficiency of the accretion disc around a Schwarzschild black hole is 10 times greater than the efficiency of the nuclear burning of hydrogen, and such disks are very strong emitters in X-rays.

Even at the temperatures of $\sim 10^7$ K that characterise an accretion disc, some heavy nuclei retain bound electrons. The small trace of iron found in the accreting matter is such a nucleus. Incident radiation from X-ray flares above and below the disc can lead to fluorescence from the highly ionised atoms in the disc; in this process an electron in the atom is de-excited from a higher energy level to a lower one and emits a photon. For iron atoms, this results in photons of energy 6.4 keV, giving a spectral line roughly in the middle of the X-ray band. As one might expect, however, the frequency of the emitted photons as measured by some observer at infinity (i.e. an astronomer on Earth) will differ from the frequency with which the photons were emitted. Qualitatively, there are two effects that cause this frequency shift. First, the photons will be gravitationally redshifted by an amount that depends on the radius from which they were emitted. Second, they will be Doppler shifted by an amount that depends on the speed and direction (relative to the distant observer) of the material from which they were emitted, in particular whether the material was moving towards or away from the observer.

Unfortunately, given the typical size of accretion disks around compact objects, and their large distance from us, the angular size of such systems as viewed from Earth is typically far smaller than the width of the observing beam of any telescope. Thus when an astronomer measures the spectrum (i.e. the photon flux as a function of frequency) of such an object, the radiation received at each frequency comes from various parts of the disc. Nevertheless, the observed spectrum is seen to consist of a much-broadened iron line, whose shape contains information about the spacetime geometry around the accreting object. In spite of the integration of contributions from across the disc, the photons coming from the inner parts of the accretion disc close to the compact object allow one to use the line profile to probe the *strong-field* regime of gravity.

As an illustration, let us calculate in some simple cases the redshift one would expect if the central object were not rotating, so that the geometry outside it is given by the Schwarzschild metric. For simplicity, take the disc to be oriented edge-on to the observer, as shown in Figure 10.8. All orbits are then in the plane of the observer and the disc, which we take to be the equatorial plane $\theta = \pi/2$.

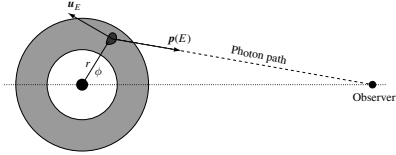


Figure 10.8 The emission of a photon by matter in an accretion disc around a compact object. The observer is viewing the disc edge-on.

The ratio of the photon's frequency at reception to that at emission is given by

$$\frac{\nu_R}{\nu_E} = \frac{\mathbf{p}(R) \cdot \mathbf{u}_R}{\mathbf{p}(E) \cdot \mathbf{u}_E} = \frac{p_\mu(R) u_R^\mu}{p_\mu(E) u_E^\mu}, \quad (10.17)$$

where $\mathbf{p}(E)$ and $\mathbf{p}(R)$ are the photon 4-momenta at emission and reception respectively, $\mathbf{u}_E$ is the 4-velocity of the material at emission and $\mathbf{u}_R$ is the 4-velocity of the observer at reception. Assuming the observer to be fixed at infinity, the components of his 4-velocity in the (t, r, θ, ϕ) coordinate system are

$$[u_R^\mu] = (1, 0, 0, 0).$$

Now consider the 4-velocity of the emitting material. Since we are assuming that this material is moving in a circular orbit it must have a 4-velocity of the form

$$[u_E^\mu] = (u_E^0, 0, 0, u_E^3).$$

Using the fact that

$$u_E^3 = \frac{d\phi}{d\tau} = \frac{d\phi}{dt} \frac{dt}{d\tau} = \frac{d\phi}{dt} u_E^0,$$

we can write the emitter's 4-velocity as

$$[u_E^\mu] = u_E^0 (1, 0, 0, \Omega),$$

where, for circular motion, $\Omega \equiv d\phi/dt = (GM/r^3)^{1/2}$, which we derived in Chapter 9. We can now fix u_E^0 by using the fact that $g_{\mu\nu} u^\mu u^\nu = c^2$. If the emitting material is at a coordinate radius r , we have

$$u_E^0 = c \left[c^2 \left(1 - \frac{2\mu}{r} \right) - r^2 \Omega^2 \right]^{-1/2} = \left(1 - \frac{3\mu}{r} \right)^{-1/2}.$$

Our general expression (10.17) therefore yields

$$\frac{\nu_R}{\nu_E} = \frac{p_0(R)}{p_0(E) u_E^0 + p_3(E) u_E^3} = \left(1 - \frac{3\mu}{r} \right)^{1/2} \frac{p_0(R)}{p_0(E)} \left[1 \pm \frac{p_3(E)}{p_0(E)} \Omega \right]^{-1},$$

where the plus sign corresponds to the emitting matter on the side of the disc moving towards the observer and the minus sign corresponds to the matter on the other side. However, the Schwarzschild metric is stationary, i.e. the metric components $g_{\mu\nu}$ do not depend explicitly on t . Thus p_0 is conserved along the geodesic, and so

$$\frac{\nu_R}{\nu_E} = \left(1 - \frac{3\mu}{r}\right)^{1/2} \left[1 \pm \frac{p_3(E)}{p_0(E)} \Omega\right]^{-1},$$

It therefore remains only to fix the ratio $p_3(E)/p_0(E)$ in order to determine the observed redshift. In general, we must use the fact that the photon worldline is null, and so $g^{\mu\nu}p_\mu p_\nu = 0$. As we are working in the equatorial plane $\theta = \pi/2$, this yields

$$\frac{1}{c^2} \left(1 - \frac{2\mu}{r}\right)^{-1} (p_0)^2 - \left(1 - \frac{2\mu}{r}\right) (p_1)^2 - \frac{1}{r^2} (p_3)^2 = 0. \quad (10.18)$$

For photons emitted from material at a general position angle ϕ , one would now need to use the geodesic equations for the photon worldline in order to eliminate p_1 . There are, however, two special cases for which this is not necessary.

The simplest case occurs when the photon is emitted from matter moving transverse to the observer, i.e. when $\phi = 0$ or $\phi = \pi$. We then have $p_3(E) = 0$, and so the observed frequency ratio is

$$\boxed{\frac{\nu_R}{\nu_E} = \left(1 - \frac{3\mu}{r}\right)^{1/2}}. \quad (10.19)$$

The other simple cases occur when the matter is moving either directly towards or away from the observer, i.e. when $\phi = -\pi/2$ or $\phi = \pi/2$. Then the radial components of the photon 4-momentum, $p_1(E)$, will be zero. From (10.18) we obtain

$$\frac{p_3(E)}{p_0(E)} = \frac{r}{c(1 - 2\mu/r)^{1/2}},$$

so that the photon frequency shift for $\phi = \mp\pi/2$ is given by

$$\boxed{\frac{\nu_R}{\nu_E} = \frac{(1 - 3\mu/r)^{1/2}}{1 \pm (r/\mu - 2)^{-1/2}}}. \quad (10.20)$$

The above discussion has been for a disc viewed edge-on. The other limiting case, when the disc is viewed face-on, is easier to analyse. Since the motion of the emitting matter is always transverse to the observer, the frequency shift is given by (10.19).

Although the observed iron line consists of photons coming from different radii in the disc, we may still calculate the smallest possible frequency (or largest redshift) present in the observed spectrum. It is clear that such photons must be emitted from the smallest possible value of r . As discussed in the previous chapter, the innermost stable circular orbit for the Schwarzschild metric is at $r = 6\mu$. Thus the smallest frequency represented is therefore given by

$$\nu_R/\nu_E = \begin{cases} \sqrt{2}/3 = 0.47 & \text{for a disc viewed edge-on,} \\ 1/\sqrt{2} = 0.71 & \text{for a disc viewed face-on.} \end{cases}$$

If the central object were rotating (so that the exterior geometry is given by the Kerr metric – see Chapter 13), then the smallest frequencies could be even lower. Figure 10.9 shows the iron spectral line measured in the galaxy MCG-6-30-15.

In general the detailed shape of the line profile depends on the mass and rotation of the central object, the inclination of the disc to the line of sight and relativistic beaming effects. It is hoped that, in the future, line profiles can be measured to

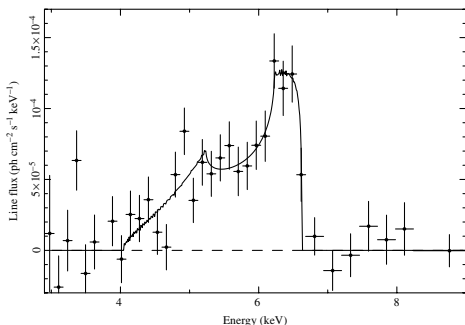


Figure 10.9 The line profile of the iron 6.4 keV spectral line from MCG-6-30-15 observed by the ASCA satellite (Y. Tanaka *et al.*, *Nature* **375**, 659, 1995). The emission line is extremely broad, the width indicating velocities of order one-third the speed of light. There is a marked asymmetry towards energies lower than the rest energy of the emission line, with a smallest energy of about 4 keV. The solid line shows a fit to the data assuming a disc around a non-rotating Schwarzschild black hole, extending between 3 and 10 Schwarzschild radii and inclined at an angle of 30° to the line of sight. Certain features suggest that the central object may in fact be rapidly rotating.

sufficient accuracy to determine the mass and angular momentum of the central compact object.

10.5 The geodesic precession of gyroscopes

We have seen how the motion of test bodies can be used to explore the geometry of a curved spacetime. If the test body has spin then the motion of its spin vector can also be used to probe the spacetime geometry. Here we discuss the idealised case of an infinitesimally small test body with spin, such as a small gyroscope.

The test body moves along a timelike geodesic curve, so its 4-velocity $u(\tau)$ is parallel-transported along its worldline. Thus, in some coordinate system, its components satisfy

$$\frac{du^\mu}{d\tau} + \Gamma^\mu_{\nu\sigma} u^\nu u^\sigma = 0.$$

Suppose that the spin of the test body is described by the 4-vector $s(\tau)$ along the geodesic. Since this vector can have no timelike component in the instantaneous rest frame of the test body, we require that at all points along the geodesic

$$s \cdot u = g_{\mu\nu} s^\mu u^\nu = 0. \quad (10.21)$$

Since the 4-velocity u of the test body is parallel-transported along its geodesic, to ensure that the inner product is conserved at all points along the worldline we require that $s(\tau)$ is also parallel-transported along the geodesic. Hence its components must satisfy

$$\frac{ds^\mu}{d\tau} + \Gamma^\mu_{\nu\sigma} s^\nu u^\sigma = 0. \quad (10.22)$$

Let us now suppose that the test body is in a circular orbit of coordinate radius r in the equatorial plane of the Schwarzschild geometry. Using the expressions we derived in Chapter 9 for the connection coefficients $\Gamma^\mu_{\nu\sigma}$ for the Schwarzschild metric in (t, r, θ, ϕ) coordinates (with $\theta = \pi/2$), one finds that most of the $\Gamma^\mu_{\nu\sigma}$ are zero. Moreover, for a test body in a circular orbit we have $u^1 = u^2 = 0$, and we find that the equations (10.22) reduce to

$$\frac{ds^0}{d\tau} + \Gamma^0_{10} s^1 u^0 = 0, \quad (10.23)$$

$$\frac{ds^1}{d\tau} + \Gamma^1_{00} s^0 u^0 + \Gamma^1_{33} s^3 u^3 = 0, \quad (10.24)$$

$$\frac{ds^2}{d\tau} = 0, \quad (10.25)$$

$$\frac{ds^3}{d\tau} + \Gamma^3_{13} s^1 u^3 = 0, \quad (10.26)$$

where the connection coefficients are given by

$$\Gamma^0_{10} = \frac{\mu}{r^2} \left(1 - \frac{2\mu}{r}\right)^{-1}, \quad \Gamma^1_{00} = \frac{\mu}{r^2} \left(1 - \frac{2\mu}{r}\right),$$

$$\Gamma^1_{33} = -r \left(1 - \frac{2\mu}{r}\right), \quad \Gamma^3_{13} = \frac{1}{r}.$$

Moreover, from our discussion in the previous section, we can write the test body's 4-velocity as

$$[u^\mu] = u^0(1, 0, 0, \Omega),$$

where $u^0 = dt/d\tau = (1 - 3\mu/r)^{-1/2}$ and $\Omega = d\phi/dt = (\mu c^2/r^3)^{1/2}$ are both constants.

Since $u^1 = u^2 = 0$ the orthogonality condition (10.21) reduces to

$$c^2(1 - 2\mu/r)s^0u^0 - r^2s^3u^3 = 0$$

and noting that $u^3/u^0 = \Omega$ we may express s^0 in terms of s^3 :

$$s^0 = \frac{\Omega r^2}{c^2(1 - 2\mu/r)} s^3.$$

Using this result it is straightforward to show that equation (10.23) is equivalent to equation (10.26). Thus the system of equations reduces to

$$\frac{ds^2}{d\tau} - \frac{r\Omega}{u^0} s^3 = 0, \quad \frac{ds^2}{d\tau} = 0, \quad \frac{ds^3}{d\tau} + \frac{u^0\Omega}{r} s^1 = 0. \quad (10.27)$$

It is more convenient to convert the τ -derivatives to t -derivatives using $u^0 = dt/d\tau$. Then, on using the third equation to eliminate s^3 from the first, the system of equations becomes

$$\frac{d^2s^1}{dt^2} + \left(\frac{\Omega}{u^0}\right)^2 s^1 = 0, \quad \frac{ds^2}{dt} = 0, \quad \frac{ds^3}{dt} + \frac{\Omega}{r} s^1 = 0.$$

Let us take the initial spatial direction $\vec{s}$ of the spin vector to be radial, so that $s^2(0) = s^3(0) = 0$. The corresponding solution to our system of equations is easily shown to be

$$s^1(t) = s^1(0) \cos \Omega' t, \quad s^2(t) = 0, \quad s^3(t) = -\frac{\Omega}{r\Omega'} s^1(0) \sin \Omega' t, \quad (10.28)$$

where $\Omega' = \Omega/u^0 = \Omega(1 - 3\mu/r)^{1/2}$. This solution shows that the spatial part $\vec{s}$ of the spin vector *rotates* relative to the radial direction with a coordinate angular speed Ω' in the negative ϕ -direction. However, the radial direction itself rotates with coordinate angular speed Ω in the positive ϕ -direction, and it is the difference between these two speeds that gives rise to the *geodesic precession*

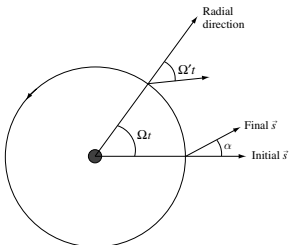


Figure 10.10 The geodesic precession effect for a spinning object in a circular orbit in the equatorial plane of the Schwarzschild geometry. Here the initial direction ($t = 0$) is radial.

effect. This is illustrated in Figure 10.10. Since one revolution is completed in a coordinate time $t = 2\pi/\Omega$, the final direction of $\vec{s}$ is therefore $2\pi + \alpha$, where $\alpha = (2\pi/\Omega)(\Omega - \Omega')$. Thus, after one revolution the spatial spin vector is rotated in the direction of the orbital motion by an angle

$$\alpha = 2\pi[1 - (1 - 3\mu/r)^{1/2}].$$

The geodesic precession effect may be observable experimentally by measuring the spacelike spin vector of a gyroscope in an orbiting spacecraft. Although the effect is small, it is cumulative. Thus, for a gyroscope in a near-Earth orbit, the precession rate is about $8''$ per year, which should be measurable. (In fact there is an additional very small effect, which may also be measurable, due to the fact that the Earth is slowly rotating and so the geometry outside it is correctly described by the Kerr metric). In April 2004, NASA launched the Gravity Probe B (GP-B) satellite to carry out this experiment and it is currently recording measurements; the results are eagerly awaited.

Exercises

10.1 Show that the equation of motion for planetary orbits in Newtonian gravity is

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{h^2},$$

where $u = 1/r$ and r is the radial distance from the centre of mass of the central object.

- 10.2 Show that the equation of motion in Exercise 10.1 has the solution

$$u = \frac{GM}{h^2}(1 + e \cos \phi),$$

and that this describes an ellipse. Show further that

$$a = \frac{h^2}{GM(1 - e^2)},$$

where $r_1 = a(1 - e)$ and $r_2 = a(1 + e)$ are the distances of closest and furthest approach respectively.

- 10.3 Verify that the general-relativistic equation of motion for planetary orbits (10.3) has the solution

$$u = \frac{GM}{h^2}(1 + e \cos \phi) + \frac{3(GM)^2}{c^2 h^4} \left[1 + e^2 \left(\frac{1}{2} - \frac{1}{6} \cos 2\phi \right) + e\phi \sin \phi \right],$$

to first order in the relativistic perturbation to the Newtonian solution.

- 10.4 Verify that the general-relativistic equation of motion for a photon trajectory (10.9) has the solution

$$u = \frac{\sin \phi}{b} + \frac{3GM}{2c^2 b^2} \left(1 + \frac{1}{3} \cos 2\phi \right),$$

to first order in the relativistic perturbation to a straight-line path.

- 10.5 Show that the gravitational deflection of light by the Sun predicted in the Newtonian theory of gravity is exactly half the value predicted in general relativity.
- 10.6 In the radar-echoes test, a photon travels from radial coordinate r to r_0 , which is the radial coordinate of the closest approach of the photon to the Sun. Verify that, to first order in μ/r , the elapsed coordinate time is given by

$$t(r, r_0) = \frac{(r^2 - r_0^2)^{1/2}}{c} + \frac{2\mu}{c} \ln \left[\frac{r + (r^2 - r_0^2)^{1/2}}{r_0} \right] + \frac{\mu}{c} \left(\frac{r - r_0}{r + r_0} \right)^{1/2}.$$

- 10.7 An accretion disc extends from $r = 6\mu$ to $r = 20\mu$ in the equatorial plane of the Schwarzschild geometry. A photon is emitted radially outwards by a particle on the inner edge of the disc and is absorbed by a particle on the outer edge of the disc. Find the ratio of the energy absorbed to that emitted.
- 10.8 For a gyroscope in a circular orbit in the equatorial plane of the Schwarzschild geometry, show that the components s^μ of its spin 4-vector satisfy equations (10.23–10.26).
- 10.9 Show that the system of equations (10.27) has the solution (10.28).
- 10.10 A gyroscope in a circular orbit of radius r in the equatorial plane of the Schwarzschild geometry has its spatial spin vector $\vec{s}$ also lying in the equatorial plane. Show that, after one complete orbit, the angle between the initial and final directions of the spatial spin vector is given by

$$\alpha = 2\pi \left[1 - (1 - 3\mu/r)^{1/2} \right],$$

irrespective of the initial direction of the spin vector. Does this still hold if the original spatial spin vector does not lie in the plane of the orbit?